

Question A — The Competing Race Between Repair and Detection

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We formalize and bound $P(\tau_{\text{ARF}} < \tau_{\text{det}})$ without any assumption on the dependence between the two clocks. Notation reminder : τ^* is the (known) drift instant ; τ_{ARF} the delay until the first tree is replaced, $\min_i \tau_i$ over M trees ; τ_{det} the delay until the first external alarm, possibly $+\infty$; F_A and F_D the cumulative distribution functions of τ_{ARF} and τ_{det} respectively ; H the post-drift observation horizon ($H = 2,000$) ; Miss the “blind spot” event, defined in Section 1.

1 Probability Space and Boundary Cases

The double-starvation case. The event $\{\tau_{\text{ARF}} = +\infty, \tau_{\text{det}} = +\infty\}$ describes a run where neither did the forest ever repair itself, nor did the alarm ever sound. This is qualitatively different from the paradox under study : the blind spot narrative relies on a *silent* repair erasing the evidence before the alarm gets its chance. If the repair itself never happened, there is nothing to erase — the model simply remains broken, with no mechanism having “cheated”. With the natural definition $\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\}$ on the extended reals, the strict inequality $\infty < \infty$ is false by convention : double starvation is therefore automatically excluded from Miss, with no special handling needed.

Strict or non-strict inequality? The stream is discrete : two processes can legitimately land on the same time step, so $P(\tau_{\text{ARF}} = \tau_{\text{det}})$ can be strictly positive — the question is not cosmetic. Physical interpretation of an equality $\tau_{\text{ARF}} = \tau_{\text{det}} = t$: the alarm sounds *exactly* when the forest repairs itself, the operator is informed at the same instant, no information is lost — unlike the true blind spot, where the alarm arrives too late or never. An equality should therefore count as a detection success.

Adopted convention.

$$\text{Miss} := \{\tau_{\text{ARF}} < \tau_{\text{det}}\} \quad (\text{strict inequality})$$

A tie is classified on the detector’s side. To be checked empirically on the trajectories : if the proportion of runs with $\tau_{\text{ARF}} = \tau_{\text{det}}$ exactly is negligible (expected, since both processes stem from underlying continuous draws), the convention choice has no material impact on the results.

2 Identification Under a Finite Horizon

Absolute censoring is the event $C_H := \{\min(\tau_{\text{ARF}}, \tau_{\text{det}}) > H\}$: neither event has happened yet at the horizon, so the winner of the race is unobservable. Conversely $\{\tau_{\text{ARF}} \leq H\}$ and $\{\tau_{\text{det}} \leq H\}$ are observable.

Four-region partition. Every run falls into exactly one of the following four cases.

Region	τ_{ARF} observed?	τ_{det} observed?	Verdict
1	$\leq H$	$\leq H$	Miss decided directly (direct comparison)
2	$\leq H$	$> H$	Certain Miss — $\tau_{\text{ARF}} \leq H < \tau_{\text{det}}$
3	$> H$	$\leq H$	Never a Miss — $\tau_{\text{det}} \leq H < \tau_{\text{ARF}}$
4	$> H$	$> H$	Unknown — this is C_H

Regions 1 and 2 together form exactly $\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}$ — a quantity that is fully computable from the data, with no assumption whatsoever about what happens beyond H .

The bound. Ignoring region 4 (lower bound), then counting it entirely as a Miss in the worst case (upper bound) :

$$\underbrace{P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss})}_{\text{computable, no assumption}} \leq P_{\text{miss}} \leq P(\{\tau_{\text{ARF}} \leq H\} \cap \text{Miss}) + P(C_H)$$

Result. The width of the bound is exactly $P(C_H)$, the proportion of doubly-censored runs : this is a Manski-type bound (partial identification). The more runs resolve before H , the tighter the interval — to be computed directly on the campaign's 2,000 (or 4,000) runs.

3 Universal Fréchet Bound

We return to the case with no finite horizon : from the marginal laws F_A and F_D alone, without any dependence assumption, we bound P_{miss} .

Lower bound. For a fixed integer s , the event $\{\tau_{\text{ARF}} \leq s < \tau_{\text{det}}\}$ always implies $\tau_{\text{ARF}} < \tau_{\text{det}}$ — it is therefore a subset of Miss, for any s . Fréchet's inequality for an intersection gives

$$P(\tau_{\text{ARF}} \leq s, \tau_{\text{det}} > s) \geq \max(0, F_A(s) - F_D(s)),$$

and since the inclusion holds for every s , we optimize by taking the largest value :

$$P_{\text{miss}} \geq \max\left(0, \sup_{s \in \mathbb{Z}} [F_A(s) - F_D(s)]\right).$$

Upper bound. We partition on $\{\tau_{\text{ARF}} \leq s\}$ and $\{\tau_{\text{ARF}} \geq s+1\}$. If $\tau_{\text{ARF}} \leq s$, the inclusion in Miss is trivial. If $\tau_{\text{ARF}} \geq s+1$ and Miss holds, then $\tau_{\text{det}} > \tau_{\text{ARF}} \geq s+1$, hence $\tau_{\text{det}} > s+1$. In both cases :

$$\text{Miss} \subseteq \{\tau_{\text{ARF}} \leq s\} \cup \{\tau_{\text{det}} > s+1\}.$$

Boole's subadditivity, followed by optimizing over s (and the trivial truncation at 1, since a probability never exceeds 1), gives

$$P_{\text{miss}} \leq \min\left(1, \inf_{s \in \mathbb{Z}} [F_A(s) + 1 - F_D(s+1)]\right).$$

Result — final bound.

$$\max\left(0, \sup_s [F_A(s) - F_D(s)]\right) \leq P_{\text{miss}} \leq \min\left(1, \inf_s [F_A(s) + 1 - F_D(s+1)]\right)$$

Valid for any dependence structure between τ_{ARF} and τ_{det} — which is what makes it a *universal* bound.

Verification on a toy case. Support $\{1, 2\}$, uniform marginals : $F_A(1) = F_D(1) = \frac{1}{2}$, $F_A(2) = F_D(2) = 1$.

Coupling	Description	$P(\text{Miss})$	Position
Comonotone	$\tau_{\text{ARF}} = \tau_{\text{det}}$ always	0	= lower bound
Independent	independent draws	0.25	strictly interior
Anti-monotone	one equals 1 when the other equals 2	0.5	= upper bound

Quick check : $\sup_s [F_A(s) - F_D(s)] = 0$ (identical marginals) gives a lower bound of 0, attained by the comonotone coupling. And $\inf_s [F_A(s) + 1 - F_D(s+1)] = 0.5$ (attained at $s = 0$ and $s = 1$) gives an upper bound of 0.5, attained by the anti-monotone coupling. The formula holds, and both bounds are attained exactly : the bound is the tightest possible on this toy case.

4 Outlook

What remains is to apply the bounds from Sections 2 and 3 to the real campaign data ($\hat{F}_A, \hat{F}_D$ estimated empirically), and to check whether the Fréchet bound — generally very wide — tightens once the known structure of the problem is used — which is exactly the point of Question B : independence conditional on the stream yields a sharper law than knowledge of the marginals alone.